

Program-7: Deploy a web application to kubernetes using minikube

Step 1: Start Minikube

minikube start

Step 2: Create nginx-deployment.yaml

nano nginx-deployment.yaml

```
GNU nano 7.2 nginx-deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 3: Apply the deployment file

kubectl apply -f nginx-deployment.yaml

Step 4: Check status of deployments and pods

kubectl get deployments

kubectl get pods

Step 5: Expose the Deployment as a Service

kubectl expose deployment nginx-deployment --type=NodePort --port=80

Step 6: Check service

kubectl get services

Step 7: Access the web app

minikube service nginx-deployment

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ minikube status
minikube
type: Control Plane
host: Running
kubernetes: Running
apiserver: Running
kubeconfig: Configured

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ nano nginx-deployment.yaml
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2            2           26s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6glkr  1/1     Running   0          36s
nginx-deployment-6f9664446b-8zcfq  1/1     Running   0          36s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP    2m
nginx-deployment    NodePort    10.108.226.27 <none>        80:30652/TCP 15s
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program8$ minikube service nginx-deployment

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30652

🔗 Opening service default/nginx-deployment in default browser...

